

Municipal Services Application — South Africa

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Overview

A C# .NET Framework 4.8 Windows Forms application that provides South African citizens with a unified platform to:

- Report municipal issues (potholes, sanitation, electricity failures, etc.)
- Browse local events and announcements with search and smart recommendations
- Track service request status using advanced data structures

Prerequisites

Requirement	Version
Operating System	Windows 10 / 11
.NET Framework	4.8
Visual Studio	2019 or later
SDK	.NET Desktop SDK (included with VS)

How to Compile

Option A — Visual Studio (recommended)

- Open Visual Studio 2019 or later.
- Click File → Open → Project/Solution.
- Navigate to the project folder and open MunicipalServicesApp.sln.
- In the Solution Explorer, right-click the project → Build.
- Confirm there are no build errors in the Output window.

Option B — Command Line (MSBuild)

```
cd path\to\MunicipalServicesApp
dotnet build MunicipalServicesApp.csproj -c Debug
```

How to Run

From Visual Studio: Press **F5** (Debug) or **Ctrl+F5** (without debugging).

From executable: `MunicipalServicesApp\bin\Debug\net48\MunicipalServicesApp.exe`

How to Use

1. Main Menu

Button	Feature	Status
Report Issues	Submit a new issue report	Active
Local Events & Announcements	Browse and search events	Active
Track Service Requests	View and search requests	Active

2. Report Issues

- Enter the location of the issue (e.g. '12 Main Street, Soweto').
- Select a category from the dropdown (Sanitation, Roads, Utilities, etc.).
- Write a description of the problem in the text box.
- Optionally click Attach File to upload an image or document (max 5 MB).
- Watch the progress bar advance as you complete each field.
- Click Submit Report — a reference ID (e.g. MSA-7841029384) is generated.
- Click Back to return to the main menu.

Engagement Feature: The progress bar advances per completed field and displays encouraging messages, motivating users to submit complete reports.

3. Local Events & Announcements

- Search — type a category or keyword and click Search to filter events.
- Clear — resets to the full sorted event list.
- Queue View — shows events in FIFO processing order.
- Search History — shows your recent searches (Stack/LIFO).
- Categories — lists all unique categories (HashSet).
- Sorted View — displays all events sorted chronologically (SortedDictionary).
- Add New Event — enter a title, category, and date to add a custom event.
- Recommendations panel (right side) — suggests related events based on your searches.

4. Track Service Requests

- Submit at least one issue via Report Issues first.
- Copy the Reference ID shown in the confirmation message.
- Open Track Service Requests.
- Paste the Reference ID in the search box and click Search (BST lookup).
- The result shows Category, Status, Priority, and submission date.
- Click Load Priority Queue to see all requests ordered by urgency.
- Click Show Workflow Flow to see the BFS graph traversal of the service pipeline.
- Click Refresh List to update the data grid.

Data Structures Used

Structure	File	Purpose
List<IssueReport>	AppData.cs	Primary storage of all submitted issues
BinarySearchTree	BinarySearchTree.cs	Fast $O(\log n)$ lookup by Reference ID
PriorityQueueManager	PriorityQueueManager.cs	Max-heap ordering issues by priority
GraphManager	GraphManager.cs	Directed graph + BFS for workflow traversal
Queue<EventItem>	EventsForm.cs	FIFO event processing order
Stack<string>	EventsForm.cs	LIFO search history
SortedDictionary	EventsForm.cs	Automatic chronological event sorting

HashSet<string>	EventsForm.cs	Unique category deduplication
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Project Structure

- MainForm.cs — Main menu dashboard
- IssueForm.cs — Report Issues form
- EventsForm.cs — Local Events & Announcements form
- EventsForm.Designer.cs — Designer stub
- StatusForm.cs — Service Request Status form
- StatusForm.Designer.cs — Designer stub
- IssueReport.cs — Issue data model
- EventItem.cs — Event data model
- AppData.cs — Shared static data store
- BinarySearchTree.cs — BST implementation
- PriorityQueueManager.cs — Priority queue (max-heap)
- GraphManager.cs — Adjacency list graph + BFS
- Program.cs — Application entry point
- MunicipalServicesApp.csproj — Project file
- MunicipalServicesApp.sln — Solution file
- README.pdf — This file

Troubleshooting

Problem	Solution
App does not compile	Ensure .NET Framework 4.8 is installed
"File too large" on attachment	Select a file under 5 MB
Grid shows no data	Submit an issue first via Report Issues
BFS shows 'Submit an issue first'	Submit at least one issue via Report Issues
Build error: missing reference	Right-click Solution → Restore NuGet Packages

References

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